

1. Requirement

You are required to develop a Minimum Viable Product (MVP) of an e-commerce web application for the use case described below. You can use any AI-assisted software engineering tool or combination of tools of your choice (e.g., GitHub Copilot, Cursor, Windsurf, Claude / Claude Code, ChatGPT, Gemini, Replit Agent, v0, Bolt, Lovable, etc.). You may also freely combine multiple tools in your workflow.

The focus of this assignment is not only the final product, but equally your workflow: how you decompose the problem, design your prompts, configure your environment, evaluate AI output, and correct or refine it. The relevant details were discussed during the lecture on the 13th July 2026.

2. Learning Outcomes

- Select and configure an AI-assisted development environment appropriate for a given task.
- Decompose a product requirement into an achievable MVP scope under time constraints.
- Apply effective prompt engineering and iterative refinement strategies during development.
- Critically evaluate AI-generated code for correctness, design quality, and maintainability.
- Document and reflect on an AI-assisted engineering workflow in a professional report.

3. The Use Case — "CeylonCart"

CeylonCart is a small online store that sells locally made products (e.g., tea, spices, handicrafts, and apparel). The owner wants a simple web storefront where customers can browse products, add them to a cart, and place an order. Real payment integration is not required. A simulated payment gateway is sufficient. All product and user data may be dummy / seeded data (hard-coded, JSON files, or a local database your choice).

3.1 Core Functional Requirements (mandatory)

#	Feature	Description
FR1	Product Catalogue	Display a list of at least 8–12 dummy products with name, image (placeholder acceptable), price, and category.
FR2	Product Details	View an individual product page with description, price, and an "Add to Cart" action.
FR3	Shopping Cart	Add / remove items, change quantities, and view a running total.
FR4	Checkout	A checkout form capturing customer name, address, and contact details (no real validation services needed).
FR5	Simulated Payment Gateway	A mock payment step (e.g., a fake card form or a "Pay Now" screen) that randomly or deterministically returns success / failure and shows an appropriate result page. No real payment provider may be used.
FR6	Order Confirmation	On successful "payment", display an order summary with a generated order ID.

3.2 Optional Extensions (attempt only if time permits)

- Search and/or category filtering of products.
- Simple user login / registration (dummy accounts acceptable).
- A basic admin view to list orders.
- Responsive / mobile-friendly styling.

Note: *A complete, working set of core requirements is worth far more than a partially working set of extensions. Scope your MVP for the 3-hour window.*

3.3 Technical Details

- Any technology stack is permitted (e.g., React/Next.js, Vue, Angular, Flask, Django, Spring Boot, Node/Express, Laravel).
- The application must run locally or on a free hosting/preview environment and be demonstrable end-to-end in your screen recording.
- No real payment gateway, real customer data, or paid third-party services may be used.
- You cannot use an existing open source shopping cart code as is. The focus is AI-Assisted software development and not Open-source library/components integration.
- The total hands-on development time must not exceed 3 hours. Keep track of your time and state it in the report.

4. What You Must Capture During Development

While building the MVP, continuously capture evidence of your process. You will need the following for your report:

1. Tools and workflow: which AI tool(s) you used, why you chose them, how they were combined, and screenshots of the tools in action at key stages.
2. Prompts and environment setup: the actual prompts (or representative examples of prompt iterations) you gave to the AI, and details of how your working environment was set up (IDE, extensions, models, project scaffolding, etc.).
3. Screen recording: a recording that shows your workflow and a demonstration of the working MVP (browsing, cart, checkout, simulated payment, confirmation). Recommended length: 5–15 minutes; edited/sped-up footage is acceptable.
4. Reflections: brief notes on challenges faced, what worked and what did not, lessons learned, and your overall reflection on AI-assisted development.

5. Deliverables and Report Structure

The final submission is a single report (PDF) containing the following sections:

Section	Content
1. Introduction	Brief description of your chosen scope and stack.
2. AI Tools & Workflow	Details and screenshots of the workflow and AI tools used, including how multiple tools were combined (if applicable).
3. Prompts & Environment Setup	The prompts used (key prompts and notable iterations) and details of your working environment configuration.

4. MVP Demonstration	Screenshots of the final MVP and the link to your screen recording (uploaded to YouTube — unlisted is fine — or Google Drive with link-sharing enabled).
5. Challenges, Learnings & Reflections	Brief notes on challenges faced, learnings, and reflections on AI-assisted development.
Appendix (optional)	Full prompt logs, additional screenshots, or a link to your source code repository.

Important: Verify that your video link is accessible before submitting (open it in an incognito/private browser window). Inaccessible links will be treated as missing deliverables.

Assessment Criteria

Criterion	What Is Assessed	Weight
Working MVP	Core requirements (FR1–FR6) demonstrated end-to-end in the recording.	35%
Workflow & Tool Usage	Clarity and sophistication of the AI-assisted workflow; evidence via screenshots.	35%
Prompts & Environment	Quality of prompts, iteration strategy, and environment documentation.	15%
Reflections	Challenges, learnings, and reflections.	15%